

Exercises and Problems for Section 14.7

Exercises

In Exercises 1–11, calculate all four second-order partial derivatives and check that $f_{xy} = f_{yx}$. Assume the variables are restricted to a domain on which the function is defined.

1. $f(x, y) = (x + y)^2$
2. $f(x, y) = (x + y)^3$
3. $f(x, y) = 3x^2y + 5xy^3$
4. $f(x, y) = e^{2xy}$
5. $f(x, y) = (x + y)e^y$
6. $f(x, y) = xe^y$
7. $f(x, y) = \sin(x/y)$
8. $f(x, y) = \sqrt{x^2 + y^2}$
9. $f(x, y) = 5x^3y^2 - 7xy^3 + 9x^2 + 11$
10. $f(x, y) = \sin(x^2 + y^2)$
11. $f(x, y) = 3 \sin 2x \cos 5y$

In Exercises 12–19, find the quadratic Taylor polynomials about $(0, 0)$ for the function.

12. $(y - 1)(x + 1)^2$
13. $(x - y + 1)^2$
14. $e^{-2x^2 - y^2}$
15. $e^x \cos y$
16. $1/(1 + 2x - y)$
17. $\cos(x + 3y)$
18. $\sin 2x + \cos y$
19. $\ln(1 + x^2 - y)$

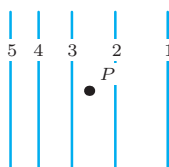
In Exercises 20–21, find the best quadratic approximation for $f(x, y)$ for (x, y) near $(0, 0)$.

20. $f(x, y) = \ln(1 + x - 2y)$
21. $f(x, y) = \sqrt{1 + 2x - y}$

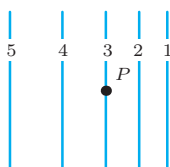
In Exercises 22–31, use the level curves of the function $z = f(x, y)$ to decide the sign (positive, negative, or zero) of each of the following partial derivatives at the point P . Assume the x - and y -axes are in the usual positions.

- (a) $f_x(P)$
- (b) $f_y(P)$
- (c) $f_{xx}(P)$
- (d) $f_{yy}(P)$
- (e) $f_{xy}(P)$

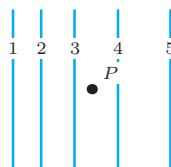
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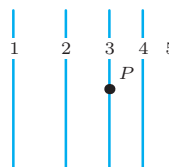
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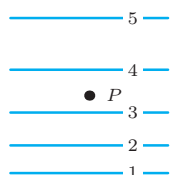
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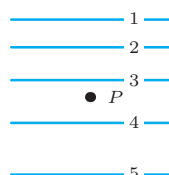
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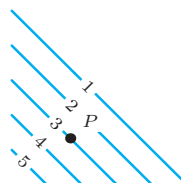
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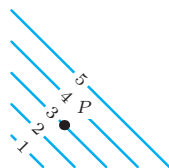
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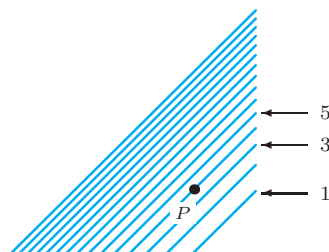
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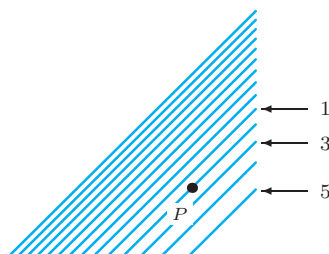
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30.



31.



Problems

In Problems 32–36, find the linear, $L(x, y)$, and quadratic, $Q(x, y)$, Taylor polynomials valid near $(1, 0)$. Compare the values of the approximations $L(0.9, 0.2)$ and $Q(0.9, 0.2)$ with the exact value of the function $f(0.9, 0.2)$.

32. $f(x, y) = \sqrt{x+2y}$ 33. $f(x, y) = x^2y$

34. $f(x, y) = xe^{-y}$

35. $F(x, y) = e^x \sin y + e^y \sin x$

36. $f(x, y) = \sin(x-1) \cos y$

In Problems 37–38, show that the function satisfies Laplace's equation, $F_{xx} + F_{yy} = 0$.

37. $F(x, y) = e^{-x} \sin y$

38. $F(x, y) = \arctan(y/x)$

39. If $u(x, t) = e^{at} \sin(bx)$ satisfies the heat equation $u_t = u_{xx}$, find the relationship between a and b .

40. (a) Check that $u(x, t)$ satisfies the heat equation $u_t = u_{xx}$ for $t > 0$ and all x , where

$$u(x, t) = \frac{1}{2\sqrt{\pi t}} e^{-x^2/(4t)}$$

(b) Graph $u(x, t)$ against x for $t = 0.01, 0.1, 1, 10$. These graphs represent the temperature in an infinitely long insulated rod that at $t = 0$ is 0°C everywhere except at the origin $x = 0$, and that is infinitely hot at $t = 0$ at the origin.

41. Figure 14.62 shows a graph of $z = f(x, y)$. Is $f_{xx}(0, 0)$ positive or negative? Is $f_{yy}(0, 0)$ positive or negative? Give reasons for your answers.

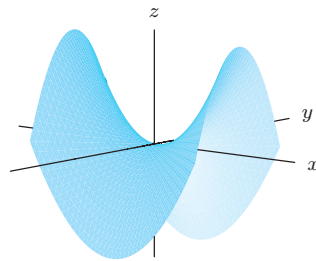


Figure 14.62

42. If $z = f(x) + yg(x)$, what can you say about z_{yy} ? Explain your answer.

43. If $z_{xy} = 4y$, what can you say about the value of
(a) z_{yx} ? (b) z_{xyx} ? (c) z_{xyy} ?

44. A contour diagram for the smooth function $z = f(x, y)$ is in Figure 14.63.

- (a) Is z an increasing or decreasing function of x ? Of y ?
- (b) Is f_x positive or negative? How about f_y ?
- (c) Is f_{xx} positive or negative? How about f_{yy} ?
- (d) Sketch the direction of $\text{grad } f$ at points P and Q .
- (e) Is $\text{grad } f$ longer at P or at Q ? How do you know?

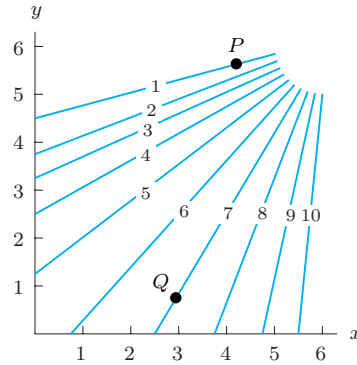


Figure 14.63

Problems 45–48 give tables of values of quadratic polynomials $P(x, y) = a + bx + cy + dx^2 + exy + fy^2$. Determine whether each of the coefficients d , e and f of the quadratic terms is positive, negative, or zero.

45.

		x		
		10	12	14
y	10	35	37	39
	15	45	47	49
	20	55	57	59

46.

		x		
		10	12	14
y	10	26	36	54
	15	31	41	59
	20	36	46	64

47.

		x		
		10	12	14
y	10	90	82	74
	15	75	87	99
	20	10	42	74

48.

		x		
		10	12	14
y	10	13	33	61
	15	28	28	36
	20	93	73	61

49. You are hiking on a level trail going due east and planning to strike off cross country up the mountain to your left. The slope up to the left is too steep now and seems to be gentler the further you go along the trail, so you decide to wait before turning off.

- (a) Sketch a topographical contour map that illustrates this story.
- (b) What information does the story give about partial derivatives? Define all variables and functions that you use.
- (c) What partial derivative influenced your decision to wait before turning?

50. The weekly production, Y , in factories that manufacture a certain item is modeled as a function of the quantity of capital, K , and quantity of labor, L , at the factory. Data shows that hiring a few extra workers increases production. Moreover, for two factories with the same number of workers, hiring a few extra workers increases production more for the factory with more capital. (With more equipment, additional labor can be used more effectively.) What does this tell you about the sign of

- (a) $\partial Y / \partial L$?
(b) $\partial^2 Y / (\partial K \partial L)$?

51. Data suggests that human surface area, S , can reasonably be modeled as a function of height, h , and weight, w . In the Dubois model, we have $\partial^2 S / \partial w^2 < 0$ and $\partial^2 S / (\partial h \partial w) > 0$. Two people A and B each gain 1 pound. Which experiences the greater increase in surface area if

- (a) They have the same weight but A is taller?
(b) They have the same height, but A is heavier?

52. Figure 14.64 shows the level curves of a function $f(x, y)$ around a maximum or minimum, M . One of the points P and Q has coordinates (x_1, y_1) and the other has coordinates (x_2, y_2) . Suppose $b > 0$ and $c > 0$. Consider the two linear approximations to f given by

$$f(x, y) \approx a + b(x - x_1) + c(y - y_1)$$

$$f(x, y) \approx k + m(x - x_2) + n(y - y_2).$$

- (a) What is the relationship between the values of a and k ?
(b) What are the coordinates of P ?
(c) Is M a maximum or a minimum?
(d) What can you say about the sign of the constants m and n ?

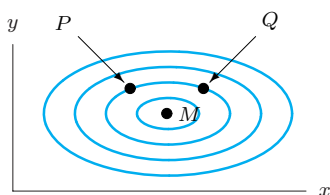


Figure 14.64

53. Consider the function $f(x, y) = (\sin x)(\sin y)$.
- (a) Find the Taylor polynomials of degree 2 for f about the points $(0, 0)$ and $(\pi/2, \pi/2)$.
(b) Use the Taylor polynomials to sketch the contours of f close to each of the points $(0, 0)$ and $(\pi/2, \pi/2)$.
54. Give an explanation of why you might expect $f_{xy}(a, b) = f_{yx}(a, b)$ using the following steps.
- (a) Write the definition of $f_x(a, b)$.
(b) Write a definition of $f_{xy}(a, b)$ as $(f_x)_y$.

- (c) Substitute for f_x in the definition of f_{xy} .
(d) Write an expression for f_{yx} similar to the one for f_{xy} you obtained in part (c).
(e) Compare your answers to parts (c) and (d). What do you have to assume to conclude that f_{xy} and f_{yx} are equal?

55. You plan to buy a used car. You are debating between a 5-year old car and a 10-year old car and thinking about the price. Experts report that the original price matters more when buying a 5-year old car than a 10-year old car. This suggests that we model the average market price, P , in dollars as a function of two variables: the original price, C , in dollars, and the age of the car, A , in years.

- (a) Give units for the following partial derivatives and say whether you think they are positive or negative. Explain your reasoning.

(a) $\partial P / \partial A$ (b) $\partial P / \partial C$

- (b) Express the experts' report in terms of partial derivatives.

- (c) Using a quadratic polynomial to model P , we have

$$P = a + bC + cA + dC^2 + eCA + fA^2.$$

Which term in this polynomial is most relevant to the experts' report?

56. The tastiness, T , of a soup depends on the volume, V , of the soup in the pot and the quantity, S , of salt in the soup. If you have more soup, you need more salt to make it taste good. Match the three stories (a)–(c) to the three statements (I)–(III) about partial derivatives.

- (a) I started adding salt to the soup in the pot. At first the taste improved, but eventually the soup became too salty and continuing to add more salt made it worse.
(b) The soup was too salty, so I started adding unsalted soup. This improved the taste at first, but eventually there was too much soup for the salt, and continuing to add unsalted soup just made it worse.
(c) The soup was too salty, so adding more salt would have made it taste worse. I added a quart of unsalted soup instead. Now it is not salty enough, but I can improve the taste by adding salt.

- (I) $\partial^2 T / \partial V^2 < 0$
(II) $\partial^2 T / \partial S^2 < 0$
(III) $\partial^2 T / \partial V \partial S > 0$

57. Let $f(x, y) = \sqrt{x + 2y + 1}$.

- (a) Compute the local linearization of f at $(0, 0)$.
(b) Compute the quadratic Taylor polynomial for f at $(0, 0)$.
(c) Compare the values of the linear and quadratic approximations in part (a) and part (b) with the true values for $f(x, y)$ at the points $(0.1, 0.1)$, $(-0.1, 0.1)$, $(0.1, -0.1)$, $(-0.1, -0.1)$. Which approximation gives the closest values?

58. Using a computer and your answer to Problem 57, draw the six contour diagrams of $f(x, y) = \sqrt{x + 2y + 1}$ and its linear and quadratic approximations, $L(x, y)$ and $Q(x, y)$, in the two windows $[-0.6, 0.6] \times [-0.6, 0.6]$ and $[-2, 2] \times [-2, 2]$. Explain the shape of the contours, their spacing, and the relationship between the contours of f , L , and Q .
59. Suppose that $f(x, y)$ has continuous partial derivatives f_x and f_y . Using the Fundamental Theorem of Calculus to evaluate the integrals, show that

$$f(a, b) = f(0, 0) + \int_{t=0}^a f_x(t, 0) dt + \int_{t=0}^b f_y(a, t) dt.$$

60. Suppose that $f(x, y)$ has continuous partial derivatives and that $f(0, 0) = 0$ and $|f_x(x, y)| \leq A$ and $|f_y(x, y)| \leq B$ for all points (x, y) in the plane. Use Problem 59 to show that $|f(x, y)| \leq A|x| + B|y|$.

This inequality shows how bounds on the partial derivatives of f limit the growth of f .

Strengthen Your Understanding

In Problems 61–62, explain what is wrong with the statement.

61. If $f(x, y) \neq 0$, then the Taylor polynomial of degree 2 approximating $f(x, y)$ near $(0, 0)$ is also nonzero.
62. There is a function $f(x, y)$ with partial derivatives $f_x = xy$ and $f_y = y^2$.

In Problems 63–65, give an example of:

63. A function $f(x, y)$ such that $f_{xx} \neq 0$, $f_{yy} \neq 0$, and $f_{xy} = 0$.
64. Formulas for two different functions $f(x, y)$ and $g(x, y)$ with the same quadratic approximation near $(0, 0)$.
65. Contour diagrams for two different functions $f(x, y)$ and $g(x, y)$ that have the same quadratic approximations near $(0, 0)$.

14.8 DIFFERENTIABILITY

In Section 14.3 we gave an informal introduction to the concept of differentiability. We called a function $f(x, y)$ *differentiable* at a point (a, b) if it is well approximated by a linear function near (a, b) . This section focuses on the precise meaning of the phrase “well approximated.” By looking at examples, we shall see that local linearity requires the existence of partial derivatives, but they do not tell the whole story. In particular, existence of partial derivatives at a point is not sufficient to guarantee local linearity at that point.

We begin by discussing the relation between continuity and differentiability. As an illustration, take a sheet of paper, crumple it into a ball and smooth it out again. Wherever there is a crease it would be difficult to approximate the surface by a plane—these are points of nondifferentiability of the function giving the height of the paper above the floor. Yet the sheet of paper models a graph which is continuous—there are no breaks. As in the case of one-variable calculus, continuity does not imply differentiability. But differentiability does *require* continuity: there cannot be linear approximations to a surface at points where there are abrupt changes in height.

Differentiability for Functions of Two Variables

For a function of two variables, as for a function of one variable, we define differentiability at a point in terms of the error and the distance from the point. If the point is (a, b) and a nearby point is $(a + h, b + k)$, the distance between them is $\sqrt{h^2 + k^2}$. (See Figure 14.65.)

A function $f(x, y)$ is **differentiable at the point** (a, b) if there is a linear function $L(x, y) = f(a, b) + m(x - a) + n(y - b)$ such that if the *error* $E(x, y)$ is defined by

$$f(x, y) = L(x, y) + E(x, y),$$

and if $h = x - a$, $k = y - b$, then the *relative error* $E(a + h, b + k)/\sqrt{h^2 + k^2}$ satisfies

$$\lim_{\substack{h \rightarrow 0 \\ k \rightarrow 0}} \frac{E(a + h, b + k)}{\sqrt{h^2 + k^2}} = 0.$$

The function f is **differentiable on a region** R if it is differentiable at each point of R . The function $L(x, y)$ is called the *local linearization* of $f(x, y)$ near (a, b) .